# K-Means Clustering — Lab Assignment

**Goal.** Implement K-Means clustering from scratch and understand how the algorithm partitions unlabeled data into $K$ clusters by minimizing intra-cluster variance. You will compute the objective, visualize convergence and cluster assignments, and automatically identify the optimal number of clusters via validation-based criteria. (No scikit-learn for the K-Means training itself.)

## Formulas

**K-Means Objective**

$$J\big(c^{(1)},\dots,c^{(m)},\mu_1,\dots,\mu_K\big) = \frac{1}{m}\sum_{i=1}^{m}\big\|x^{(i)}-\mu_{c^{(i)}}\big\|^2$$

where $c^{(i)}$ is the index of the centroid assigned to $x^{(i)}$.

**Centroid Update**

$$\mu_k = \frac{\sum_{i=1}^{m}\mathbf{1}\{c^{(i)}=k\}\,x^{(i)}}{\sum_{i=1}^{m}\mathbf{1}\{c^{(i)}=k\}}$$

**Within-Cluster Sum of Squares (WCSS)**

$$\mathrm{WCSS}(K) = \sum_{k=1}^{K}\sum_{x^{(i)}\in C_k}\big\|x^{(i)}-\mu_k\big\|^2$$

**Silhouette Coefficient**

$s(i) = \dfrac{b(i)-a(i)}{\max\{a(i),\,b(i)\}}$, $\quad a(i)$ = mean distance to points in same cluster, $b(i) = \min\limits_{\ell\neq c^{(i)}}$ mean distance to cluste

## Exercise 2 — Automatic Selection of Optimal $K$ (Elbow Method Only)

**Objective.** To automatically determine the optimal number of clusters $K$ in K-Means clustering using the *Elbow Method*, evaluated on a validation set.

**Task.**

(a) For each $K \in \{1,2,\dots,K_{\max}\}$:

- Train the K-Means algorithm on the `train.csv` data using *k-means++* initialization.
- Assign the validation data (`val.csv`) to the nearest centroids and compute the **Within-Cluster Sum of Squares (WCSS)**:

$$\mathrm{WCSS}(K) = \sum_{i=1}^{m_{\mathrm{val}}}\big\|x^{(i)}-\mu_{c(i)}\big\|^2$$

where $\mu_{c(i)}$ denotes the centroid of the cluster assigned to $x^{(i)}$.

(b) Determine the **Elbow Point** $K_{\text{elbow}}$ using the *kneedle* rule: Let the points $(K, \text{WCSS}(K))$ for $K = 1, \ldots, K_{\max}$ define a curve. The elbow is identified as the $K$ having the maximum perpendicular distance from the straight line connecting $(1, \text{WCSS}(1))$ and $(K_{\max}, \text{WCSS}(K_{\max}))$:

$$K_{\text{elbow}} = \arg\max_K d_K$$

where

$$d_K = \frac{|(K_{\max} - 1)(\text{WCSS}_1 - \text{WCSS}_K) - (1 - K)(\text{WCSS}_1 - \text{WCSS}_{K_{\max}})|}{\sqrt{(K_{\max} - 1)^2 + (\text{WCSS}_1 - \text{WCSS}_{K_{\max}})^2}}.$$

(c) Retrain K-Means on the training data using $K = K_{\text{elbow}}$ and evaluate on the test data (`test.csv`). Compute and report:

$$J_{\text{test}} = \frac{1}{m_{\text{test}}} \sum_{i=1}^{m_{\text{test}}} \|x^{(i)} - \mu_{c(i)}\|^2, \quad \text{WCSS}_{\text{test}} = \sum_{i=1}^{m_{\text{test}}} \|x^{(i)} - \mu_{c(i)}\|^2.$$

**Input Format.**

- Two lines are provided via standard input:

  ```
  Kmax=<int>
  iterations=<int>
  ```

- Three files are present in the working directory:
  - `train.csv` — training data
  - `val.csv` — validation data
  - `test.csv` — test data

**Output Format.** The program must print exactly three lines to standard output (values rounded to two decimals):

```
Elbow_K=<int>
WCSS_Val=[W1,W2,...,WKmax]
Test_J_Elbow=<float>  Test_WCSS_Elbow=<float>
```

where

- `Elbow_K` is the optimal $K_{\text{elbow}}$.
- `WCSS_Val` is the validation WCSS list for $K = 1, \ldots, K_{\max}$.
- `Test_J_Elbow` and `Test_WCSS_Elbow` are computed on the test set.

**Constraints.**

- $1 \leq K_{\max} \leq \min(12, m_{\text{train}})$.
- `iterations` $\in [1, 100]$.
- Only `numpy` and `matplotlib` may be used. Do not import or call `sklearn.cluster.KMeans`.

**Expected Behavior.**

(i) Compute $\text{WCSS}(K)$ on validation data for all $K$.

(ii) Select $K_{\text{elbow}}$ via the kneedle rule.

(iii) Retrain and compute $J_{\text{test}}$ and $\text{WCSS}_{\text{test}}$.

(iv) Save the optional plot `elbow_curve.png`.

**Sample Input.**

```
Kmax=6
iterations=10
```

**Sample Output.**

```
Elbow_K=3
WCSS_Val=[1379.26,638.25,106.82,94.85,80.87,70.47]
Test_J_Elbow=1.24  Test_WCSS_Elbow=111.57
```

**Evaluation Criteria (for VPL/Moodle).**

- Correct computation of WCSS($K$) values.
- Proper identification of $K_{\text{elbow}}$.
- Accuracy of $J_{\text{test}}$ and $\text{WCSS}_{\text{test}}$ within a small numeric tolerance ($\pm 0.05$).
- Output format matches exactly as specified.

**Implementation Hints.**

- Use *k-means++* initialization with a fixed random seed (e.g., 0) for reproducibility.
- Handle empty clusters by reseeding the centroid with a random training point.
- Ensure convergence detection based on unchanged cluster assignments.
- When computing squared distances, clip small negative values to 0 to avoid numeric errors.